

Analysis of Breast Tumor Tissue Biopsies Using PCA

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Introduction

Data consulting & education Pills_R. Training Material

[website] <https://agustincastro.es>

[RPods] <https://rpubs.com/acastro>

[GitHub-R] <https://github.com/acastromartinez/GITHUB---R>

About the Technique

In this practice, we will work with the **Principal Component Analysis (PCA) technique** to reduce dimensionality in datasets with a large number of variables. PCA simplifies the analysis and enhances the visualization and interpretation of results by transforming the original variables into a new set of principal components. These components retain the maximum amount of information possible using fewer dimensions. The first principal component explains the most variance in the data, while each subsequent component captures the maximum remaining variance without overlapping the information already explained by the previous components.

PCA is particularly valuable for large datasets, as it allows for significant simplification of the analysis and effective visualization without sacrificing relevant information.

About the Dataset

We will use the dataset **biopsy** from the **MASS** package, one of the most popular and widely used in statistical analysis. It was developed by the group of Venables and Ripley and accompanies the book **Modern Applied Statistics with S**.

The dataset **biopsy** contains information for the **classification of breast tumors** as **benign** or **malignant** based on characteristics obtained from fine needle aspiration biopsy images of breast masses. The data includes both quantitative features of the cells and the final tumor classification. It is a small, manageable dataset that offers a variety of features for applying multiple modeling techniques and is ideal for teaching and experimenting with data analysis and machine learning. Additionally, it is well-documented and supported in the MASS package for R.

Packages

```
library(MASS)
# install.packages("MASS") if necessary

library(tidyverse)
library(ggplot2)
library(corrplot) ##
library(GGally) ##
library(factoextra) ##
library(gridExtra) ##

## ver anexo
```

Load dataset information

Data obtained from the **University of Wisconsin Hospitals, Madison (Dr. Wolberg)**. It is based on the evaluation of 699 breast tumor biopsies. Each of the nine attributes V1-V9 is scored on a scale from 1 to 10. The two-level classification, **benign** and **malignant**, is also known.

```
data(biopsy)
citation("MASS") # how to cite the dataset

## To cite the MASS package in publications use:
##
## Venables, W. N. & Ripley, B. D. (2002) Modern Applied Statistics with
## S. Fourth Edition. Springer, New York. ISBN 0-387-95457-0
##
## A BibTeX entry for LaTeX users is
##
## @Book{,
##   title = {Modern Applied Statistics with S},
##   author = {W. N. Venables and B. D. Ripley},
##   publisher = {Springer},
##   edition = {Fourth},
##   address = {New York},
##   year = {2002},
##   note = {ISBN 0-387-95457-0},
##   url = {https://www.stats.ox.ac.uk/pub/MASS4/},
## }
```

Data review and cleaning

```
dim(biopsy) # dimensions (dataset with 99 rows x 11 variables)
```

```
## [1] 699 11
```

```
str(biopsy) # types of variables (see appendix)
```

```
## 'data.frame':    699 obs. of  11 variables:
## $ ID      : chr  "1000025" "1002945" "1015425" "1016277" ...
## $ V1      : int   5 5 3 6 4 8 1 2 2 4 ...
## $ V2      : int   1 4 1 8 1 10 1 1 1 2 ...
## $ V3      : int   1 4 1 8 1 10 1 2 1 1 ...
## $ V4      : int   1 5 1 1 3 8 1 1 1 1 ...
## $ V5      : int   2 7 2 3 2 7 2 2 2 2 ...
## $ V6      : int   1 10 2 4 1 10 10 1 1 1 ...
## $ V7      : int   3 3 3 3 3 9 3 3 1 2 ...
## $ V8      : int   1 2 1 7 1 7 1 1 1 1 ...
## $ V9      : int   1 1 1 1 1 1 1 1 5 1 ...
## $ class: Factor w/ 2 levels "benign","malignant": 1 1 1 1 1 2 1 1 1 1 ...
```

```
head(biopsy)
```

```
##      ID V1 V2 V3 V4 V5 V6 V7 V8 V9      class
## 1 1000025 5 1 1 1 2 1 3 1 1    benign
## 2 1002945 5 4 4 5 7 10 3 2 1    benign
## 3 1015425 3 1 1 1 2 2 3 1 1    benign
## 4 1016277 6 8 8 1 3 4 3 7 1    benign
## 5 1017023 4 1 1 3 2 1 3 1 1    benign
## 6 1017122 8 10 10 8 7 10 9 7 1 malignant
```

Missing cases, NAs

We check for missing data in the dataset and find that there are 16 NAs in the variable V6. The **grid.table** function is not included by default in R. It requires the installation of the **gridExtra** library.

```
colSums(is.na(biopsy)) # there are 16 NAs in column V6
```

```
##      ID      V1      V2      V3      V4      V5      V6      V7      V8      V9 class
##      0         0         0         0         0         0      16         0         0         0      0
```

```
grid.table(biopsy[!complete.cases(biopsy),])
```

	ID	V1	V2	V3	V4	V5	V6	V7	V8	V9	class
24	1057013	8	4	5	1	2	NA	7	3	1	malignant
41	1096800	6	6	6	9	6	NA	7	8	1	benign
140	1183246	1	1	1	1	1	NA	2	1	1	benign
146	1184840	1	1	3	1	2	NA	2	1	1	benign
159	1193683	1	1	2	1	3	NA	1	1	1	benign
165	1197510	5	1	1	1	2	NA	3	1	1	benign
236	1241232	3	1	4	1	2	NA	3	1	1	benign
250	169356	3	1	1	1	2	NA	3	1	1	benign
276	432809	3	1	3	1	2	NA	2	1	1	benign
293	563649	8	8	8	1	2	NA	6	10	1	malignant
295	606140	1	1	1	1	2	NA	2	1	1	benign
298	61634	5	4	3	1	2	NA	2	3	1	benign
316	704168	4	6	5	6	7	NA	4	9	1	benign
322	733639	3	1	1	1	2	NA	3	1	1	benign
412	1238464	1	1	1	1	1	NA	2	1	1	benign
618	1057067	1	1	1	1	1	NA	1	1	1	benign

We remove the rows that contain missing data in variable V6. The final dimension of **biopsy** is 683 rows x 11 columns (vars).

```
biopsy <- drop_na(biopsy, V6) # removing NAs
sum(is.na(biopsy$V6)) # another way to check
```

```
## [1] 0
```

```
dim(biopsy)
```

```
## [1] 683 11
```

Duplicate IDs

There are samples with duplicate IDs but with different measured values for the various variables. Therefore, we do not remove any of these duplicate IDs, as they contain different information.

```
# we check if there are duplicate ID samples in the dataset
sum(duplicated(biopsy$ID)) # there are 53 duplicate IDs
```

```
## [1] 53
```

```
biopsy$ID[duplicated(biopsy$ID)] # these are the duplicate IDs
```

```
## [1] "1033078" "1070935" "1143978" "1171710" "1173347" "1174057" "1212422"
## [8] "1218860" "1017023" "1100524" "1116116" "1168736" "1182404" "1182404"
## [15] "1198641" "1182404" "1198641" "320675" "704097" "493452" "560680"
## [22] "1114570" "1158247" "1276091" "1276091" "1276091" "1293439" "734111"
## [29] "1182404" "1276091" "1105524" "1115293" "1182404" "1320077" "769612"
## [36] "798429" "1116192" "1240603" "1299924" "1321942" "385103" "411453"
## [43] "822829" "1061990" "1238777" "1277792" "1299596" "1339781" "1354840"
## [50] "466906" "654546" "695091" "897471"
```

```
duplicated <- biopsy$ID[duplicated(biopsy$ID)]
table(duplicated) # number of times each ID is repeated
```

```
## duplicated
## 1017023 1033078 1061990 1070935 1100524 1105524 1114570 1115293 1116116 1116192
##      1      1      1      1      1      1      1      1      1      1
## 1143978 1158247 1168736 1171710 1173347 1174057 1182404 1198641 1212422 1218860
##      1      1      1      1      1      1      5      2      1      1
## 1238777 1240603 1276091 1277792 1293439 1299596 1299924 1320077 1321942 1339781
##      1      1      4      1      1      1      1      1      1      1
## 1354840 320675 385103 411453 466906 493452 560680 654546 695091 704097
##      1      1      1      1      1      1      1      1      1      1
## 734111 769612 798429 822829 897471
##      1      1      1      1      1
```

```
# we examine the values of the variables for the different samples with duplicate IDs
# they are different, so we do not remove duplicates.
```

```
biopsy_duplicates <- biopsy %>%
  filter(ID %in% ID[duplicated(ID)]) %>%
  arrange(ID)
head(biopsy_duplicates)
```

```
##      ID V1 V2 V3 V4 V5 V6 V7 V8 V9 class
## 1 1017023 4 1 1 3 2 1 3 1 1 benign
## 2 1017023 6 3 3 5 3 10 3 5 3 benign
## 3 1033078 2 1 1 1 2 1 1 1 5 benign
## 4 1033078 4 2 1 1 2 1 2 1 1 benign
## 5 1061990 1 1 3 2 2 1 3 1 1 benign
## 6 1061990 4 1 1 1 2 1 2 1 1 benign
```

Benign vs Malignant

Number of cases in the dataset for each level of the ‘class’ factor. Tumor masses are classified into one group or the other.”

```
table(biopsy$class)
```

```
##
##      benign malignant
##      444      239
```

Correlation between variables

To study the correlation between the variables used for tumor classification, we need to create a correlation matrix. To do this, we create a subset from **biopsy** that contains exclusively the data for the variables of interest (V1 to V9), removing the ID column. We will name this new dataframe **biopsy_vars**, as it only contains the mentioned variables.

```
biopsy_vars <- biopsy[, -c(1, 11)]
str(biopsy_vars)
```

```
## 'data.frame':    683 obs. of  9 variables:
## $ V1: int  5 5 3 6 4 8 1 2 2 4 ...
## $ V2: int  1 4 1 8 1 10 1 1 1 2 ...
## $ V3: int  1 4 1 8 1 10 1 2 1 1 ...
## $ V4: int  1 5 1 1 3 8 1 1 1 1 ...
## $ V5: int  2 7 2 3 2 7 2 2 2 2 ...
## $ V6: int  1 10 2 4 1 10 10 1 1 1 ...
## $ V7: int  3 3 3 3 3 9 3 3 1 2 ...
## $ V8: int  1 2 1 7 1 7 1 1 1 1 ...
## $ V9: int  1 1 1 1 1 1 1 1 5 1 ...
```

Now we create the correlation matrix using R's `cor` function. We will store it in **biopsy_vars_cor**.

```
biopsy_vars_cor <- cor(biopsy_vars)
print(biopsy_vars_cor)
```

```
##           V1           V2           V3           V4           V5           V6           V7
## V1 1.0000000 0.6424815 0.6534700 0.4878287 0.5235960 0.5930914 0.5537424
## V2 0.6424815 1.0000000 0.9072282 0.7069770 0.7535440 0.6917088 0.7555592
## V3 0.6534700 0.9072282 1.0000000 0.6859481 0.7224624 0.7138775 0.7353435
## V4 0.4878287 0.7069770 0.6859481 1.0000000 0.5945478 0.6706483 0.6685671
## V5 0.5235960 0.7535440 0.7224624 0.5945478 1.0000000 0.5857161 0.6181279
## V6 0.5930914 0.6917088 0.7138775 0.6706483 0.5857161 1.0000000 0.6806149
## V7 0.5537424 0.7555592 0.7353435 0.6685671 0.6181279 0.6806149 1.0000000
## V8 0.5340659 0.7193460 0.7179634 0.6031211 0.6289264 0.5842802 0.6656015
## V9 0.3509572 0.4607547 0.4412576 0.4188983 0.4805833 0.3392104 0.3460109
##           V8           V9
## V1 0.5340659 0.3509572
## V2 0.7193460 0.4607547
## V3 0.7179634 0.4412576
## V4 0.6031211 0.4188983
## V5 0.6289264 0.4805833
## V6 0.5842802 0.3392104
## V7 0.6656015 0.3460109
## V8 1.0000000 0.4337573
## V9 0.4337573 1.0000000
```

We create the significance matrix with the p-values obtained from the **correlation matrix** calculation. To do this, we use the R function **cor.mtest** (in this case, with a 0.95 confidence level).

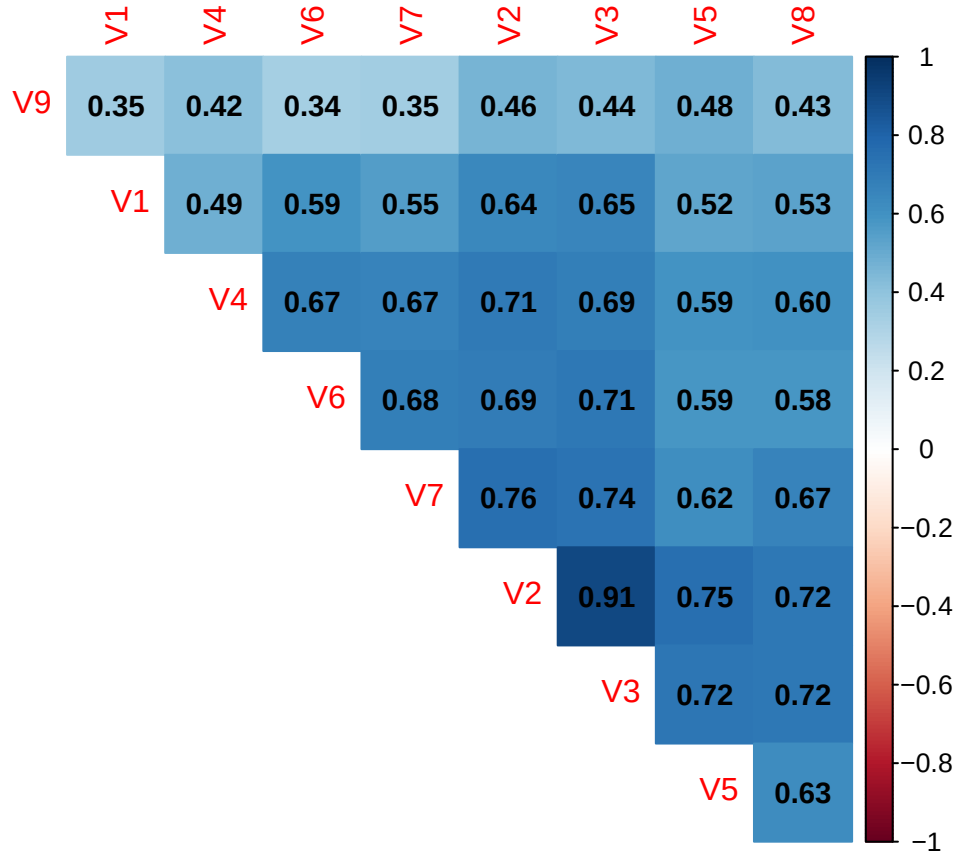
```
biopsy_vars_sig <- cor.mtest(biopsy_vars, conf.level = 0.95)
print(biopsy_vars_sig$p)
```

```
##           V1           V2           V3           V4           V5
## V1 0.000000e+00 8.964173e-81 2.064616e-84 4.027956e-42 2.411759e-49
## V2 8.964173e-81 0.000000e+00 2.567742e-258 1.544338e-104 3.535863e-126
## V3 2.064616e-84 2.567742e-258 0.000000e+00 4.146228e-96 3.061101e-111
## V4 4.027956e-42 1.544338e-104 4.146228e-96 0.000000e+00 1.627087e-66
## V5 2.411759e-49 3.535863e-126 3.061101e-111 1.627087e-66 0.000000e+00
## V6 4.050902e-66 2.402961e-98 1.807287e-107 2.058229e-90 3.828707e-64
## V7 3.880813e-56 3.184894e-127 3.567289e-117 1.153511e-89 3.199163e-73
## V8 1.260114e-51 7.433029e-110 3.018221e-109 6.883291e-69 1.734754e-76
## V9 3.148289e-21 3.398097e-37 6.531371e-34 2.125287e-30 9.266128e-41
##           V6           V7           V8           V9
## V1 4.050902e-66 3.880813e-56 1.260114e-51 3.148289e-21
## V2 2.402961e-98 3.184894e-127 7.433029e-110 3.398097e-37
## V3 1.807287e-107 3.567289e-117 3.018221e-109 6.531371e-34
## V4 2.058229e-90 1.153511e-89 6.883291e-69 2.125287e-30
## V5 3.828707e-64 3.199163e-73 1.734754e-76 9.266128e-41
## V6 0.000000e+00 4.391890e-94 9.158074e-64 7.473326e-20
## V7 4.391890e-94 0.000000e+00 1.312645e-88 1.214400e-20
## V8 9.158074e-64 1.312645e-88 0.000000e+00 1.053441e-32
## V9 7.473326e-20 1.214400e-20 1.053441e-32 0.000000e+00
```

Using the **corrplot** function, we create a **correlation plot**. Here, we can see that the most correlated variables ($r = 0.91$) are **V2** and **V3** (cell size and shape uniformity, respectively). Several variables are correlated with r values close to 0.8. All r values obtained in the matrix are positive (+) and significant with a 95% confidence level. **The use of the corrplot function requires the installation of the corrplot library, as it is not included by default in R.**

[Note] PCA helps simplify the analysis by reducing the number of variables while preserving most of the variance or relevant information in the data. This is particularly useful when the variables are highly correlated, as PCA can identify principal components that capture the underlying relationships between variables and reduce redundancy. Additionally, it makes interpretation easier by reducing the number of individual variables to consider.

```
corrplot(biopsy_vars_cor,
         p.mat = biopsy_vars_sig$p, sig.level = 0.05,
         method = "color",
         order = "hclust",
         type = "upper",
         diag = FALSE,
         addCoef.col = "black",
         number.cex = 0.9)
```



PCA

The **prcomp** function in R is used to perform Principal Component Analysis (PCA). In this case, **center = TRUE** ensures that the variables are centered around zero mean, eliminating the bias introduced by different variable means, and **scale = TRUE** normalizes the variables to have unit variance, ensuring that all variables contribute equally to the analysis, regardless of their original scales.

```
biopsy_PCA <- prcomp(biopsy_vars, center = TRUE, scale = TRUE)
```

We can use **prcomp** on the original dataframe **biopsy** (where we did not remove the ID variable of type char). In this case, it is necessary to specify the columns/variables that will be considered in the analysis.

```
biopsy_PCA <- prcomp(biopsy[, -c(1, 11)], center = TRUE, scale = TRUE)
```

Scree plot

A **Scree Plot** is a graph that shows the variance explained by each principal component in PCA. The x-axis represents the principal components (sorted according to the amount of variance they explain), and the y-axis shows the variance explained by each component.

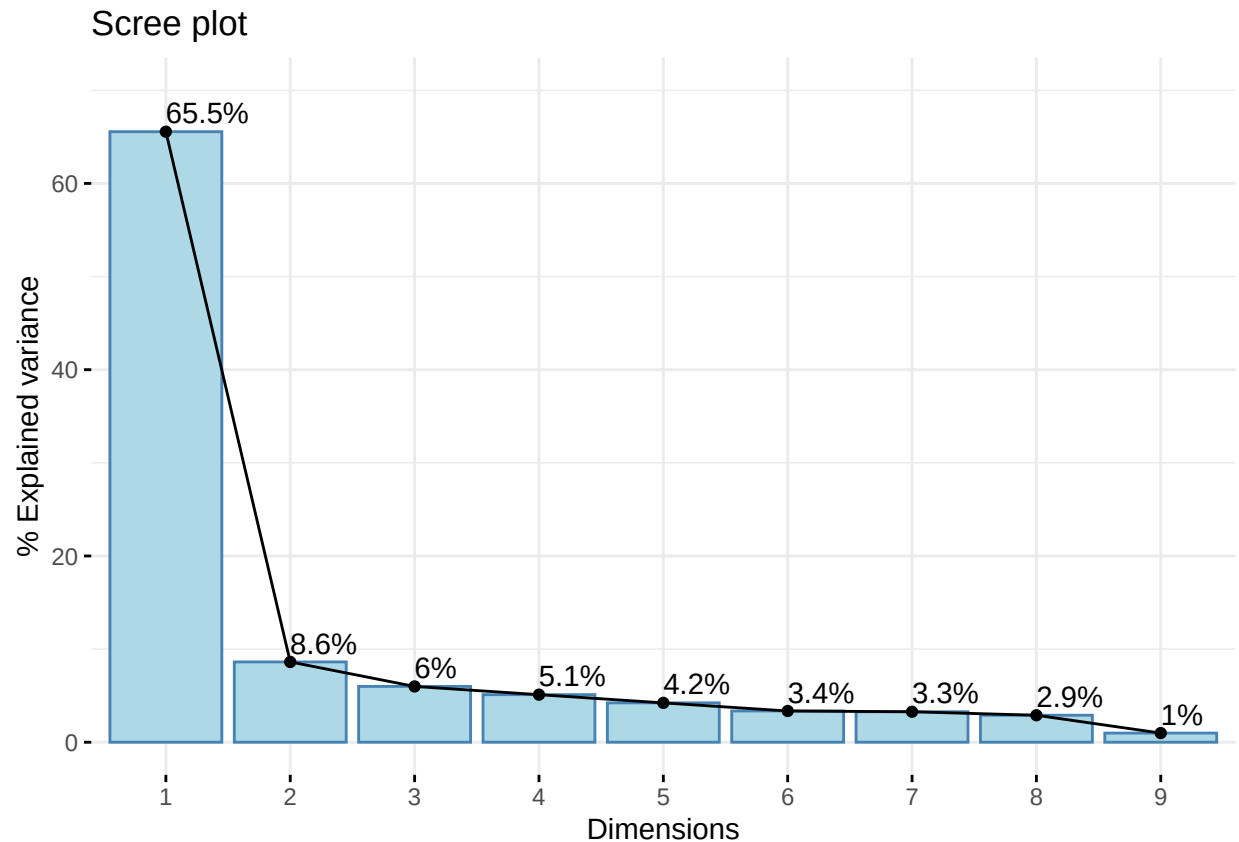
In the plot, the 'elbow' or inflection point is of particular importance. It is where the slope of the graph changes significantly. By observing this, we can determine the number of principal components that explain a significant amount of variance.

```

scree_plot <- fviz_screplot(biopsy_PCA,
  title = "Scree plot",
  xlab = "Dimensions",
  ylab = "% Explained variance",
  barfill = "lightblue",
  ylim = c(0, 70),
  addlabels = TRUE)

scree_plot

```



In PCA, the first components capture the majority of the variability in the data. In this specific case, our first component explains the largest portion of the variance (65.5%), while the second explains a smaller proportion (8.6%). **These two components together explain 74.1% of the total variance.** The significant difference between the first and second components indicates that the first is clearly more significant for describing the structure of the data we are analyzing. However, the second component may also be useful for informational purposes. This is something we will explore further as we delve into the results obtained from the PCA.

Contribution of the variables

In the context of PCA, the **contribution of the variables** refers to how much each original variable contributes to a specific principal component. This concept is key to understanding how the original variables combine to form the principal components, as well as interpreting the importance of each variable in the reduced data space representation.

By taking a closer look, we might draw some conclusions.

In order, the variables that contribute the most are: **V2** (uniformity of cell size): **Uniformity of cell size**. **V3** (uniformity of cell shape): **Uniformity of cell shape**. **V7** (bland chromatin): **Bland chromatin**. **V5** (single epithelial cell size): **Size of individual epithelial cells**. **V8** (normal nucleoli): **Normal nucleoli**. **V6** (bare nuclei): **Presence of bare nuclei**.

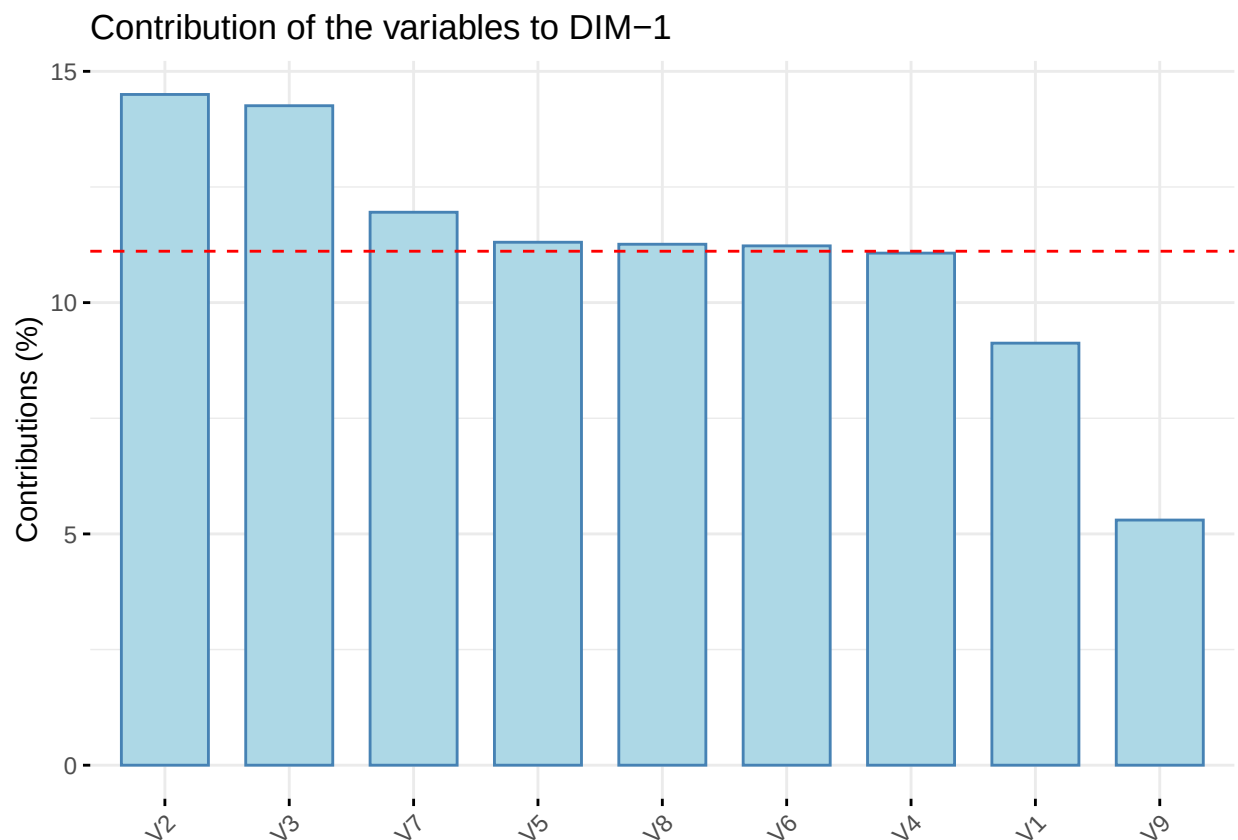
The red line represents a threshold indicating the value of contribution **above which variables are considered significant for the principal component we are analyzing**.

The first principal component would be closely associated with characteristics related to the morphology and size of the breast tissue cells.

Furthermore, the relationship between **bland chromatin (V7)** and the morphological characteristics of cells, including their size and shape, is supported by scientific research in cell biology and cancer. There is evidence linking bland chromatin with breast cancer due to its influence on DNA accessibility and gene expression regulation (in this case, increased transcriptional activity). Alterations in the structure and epigenetic modifications of chromatin can facilitate the activation of genes associated with malignancy and contribute to tumor progression.

```
contrib_plot_dim1 <- fviz_contrib(biopsy_PCA,
                                choice = "var",
                                axes = 1,
                                fill = "lightblue")

contrib_plot_dim1 +
  ggtitle("Contribution of the variables to DIM-1") +
  ylab("Contributions (%)")
```



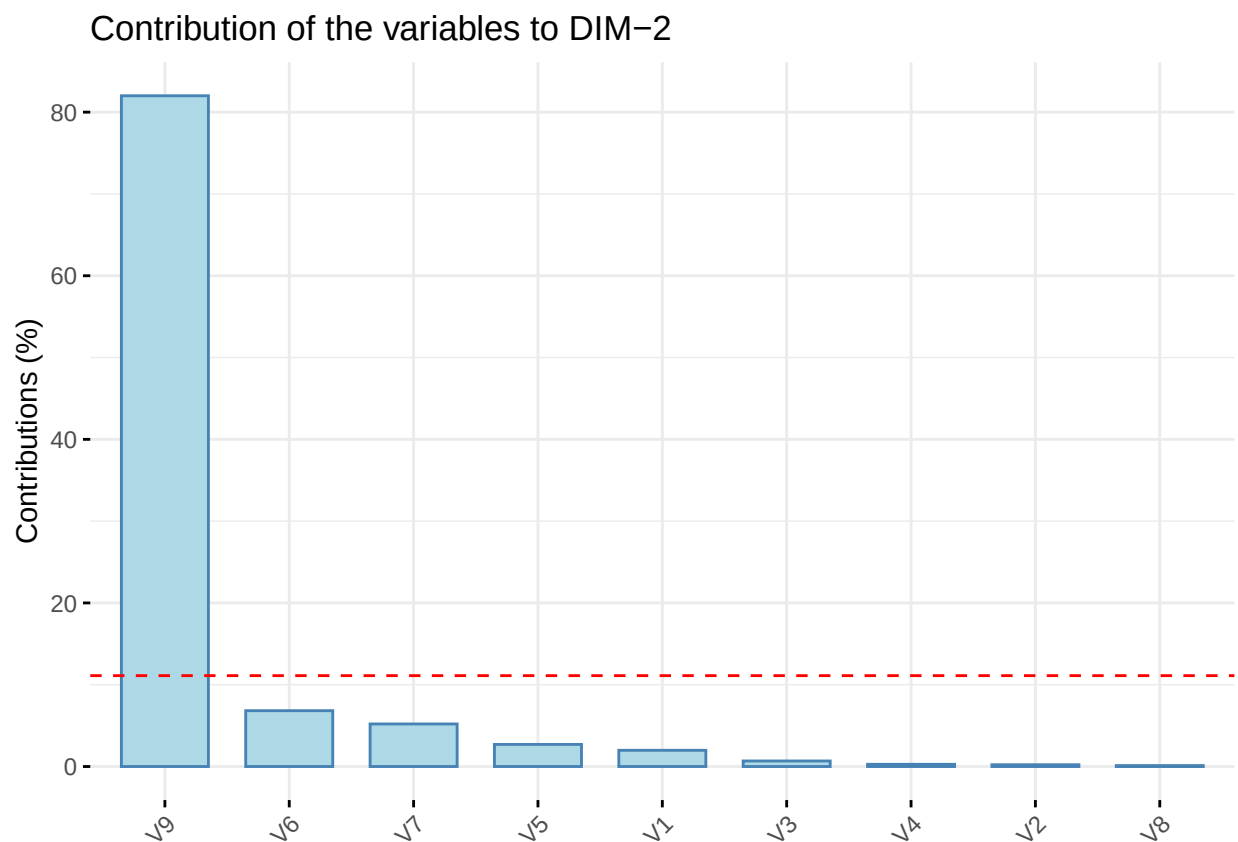
In the case of the **second principal component**, we have a clear dominance of a single variable:

V9 (mitoses): Number of mitoses.

This second component would reflect **cellular activity in terms of proliferation**. Tissue samples with non-cancerous cells would show normal mitosis rates, whereas those from malignant tumors would be altered, with significantly higher proliferation values.

```
contrib_plot_dim2 <- fviz_contrib(biopsy_PCA,
                                  choice = "var",
                                  axes = 2,
                                  fill = "lightblue")

contrib_plot_dim2 +
  ggtitle("Contribution of the variables to DIM-2") +
  ylab("Contributions (%)")
```

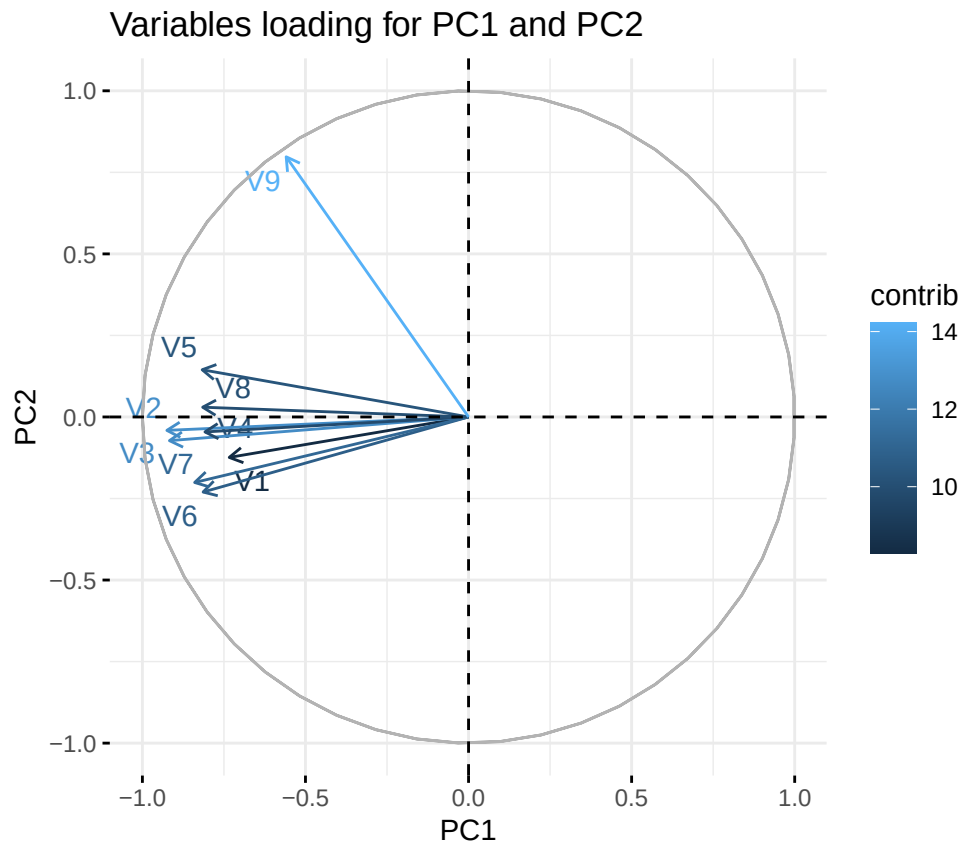


Variables loading

The ‘**variable loadings**’ indicates how **each original variable influences the principal components**. The loadings show how important each variable is for each component, helping to understand how the principal components are formed and which variables are most relevant in explaining the variability of the data.

We can see how the variables **V2** and **V3** have a very close and strong relationship with PC1 (principal component 1), while **V9** is clearly more associated with PC2.

```
fviz_pca_var(biopsy_PCA, col.var = "contrib", repel = TRUE, axes = c(1, 2)) +
labs(title="Variables loading for PC1 and PC2", x="PC1", y="PC2")
```

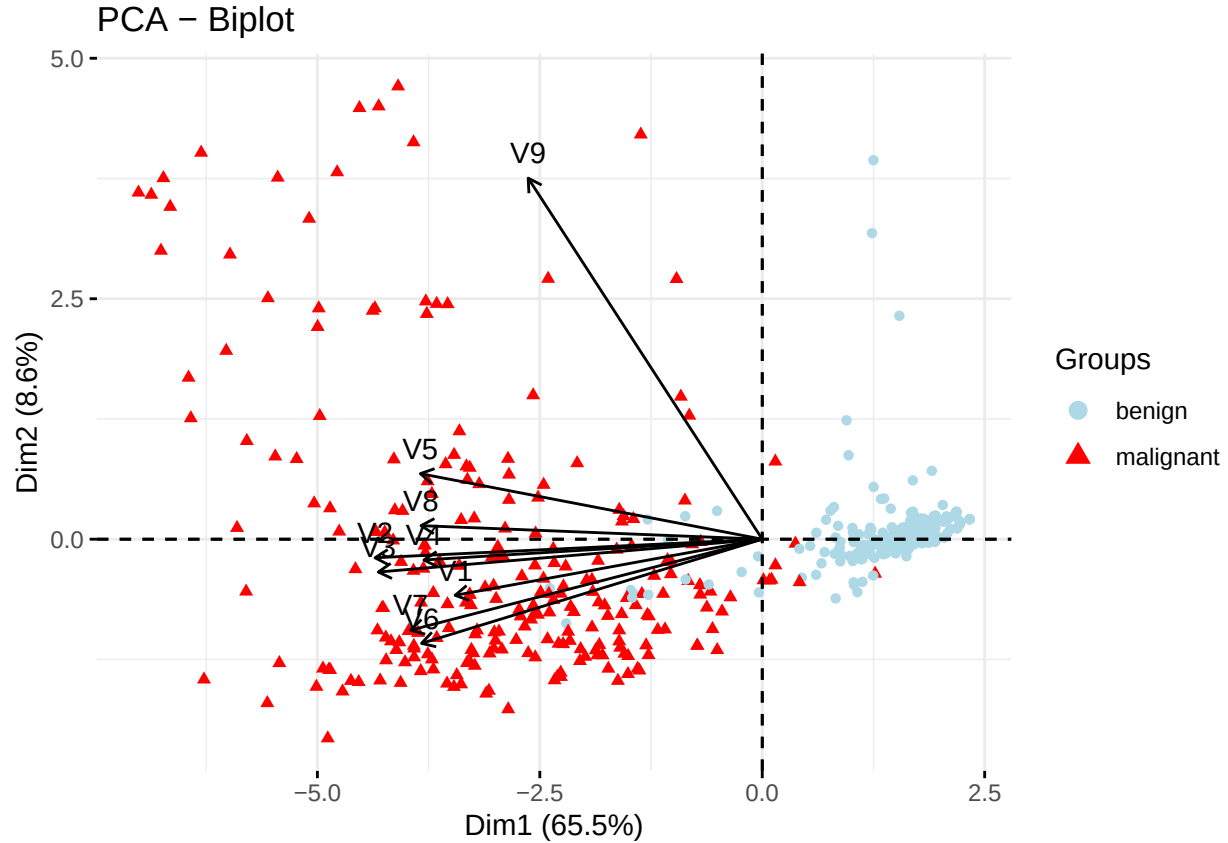


BIPLOT

A biplot is a graph that combines the representation of observations and variables in the same space, typically in the context of Principal Component Analysis (PCA). **Observations** are represented as points on the plot, showing how the data are distributed based on the principal components. **Variables** are represented as arrows or vectors, indicating the direction and magnitude of their influence on the principal components.

In this case, we represent the observations of the 'class' variable, which in the **biopsy_PCA** dataframe contains information about the **tumor classification** ('benign' or 'malignant'). To facilitate the visualization of the result in the biplot, it is also possible to use different colors to indicate the 'benign' and 'malignant' categories.

```
fviz_pca_biplot(biopsy_PCA,
  label = "var",
  habillage = biopsy$class,
  axes = c(1,2),
  addEllipse = FALSE,
  col.var = "black") +
scale_color_manual(values = c("lightblue", "red"))
```



Conclusions

We started with a dataframe containing data from 9 variables (V1 to V9) corresponding to samples obtained from breast tumor tissue biopsies.

- For each sample, the tumor is classified as ‘benign’ or ‘malignant’.
- PCA allows us to reduce the dimensionality to **one** or **two factors** (principal components), achieving nearly 80% of the total variance explanation (specifically, **74.1%**).
- The first principal component **PC1** is closely associated with the **cellular morphology** of the sampled breast tissue (uniformity in cell shape and size).
- The second principal component **PC2** is related to cellular proliferation, reflecting the different rates of **cell proliferation** (mitosis) between benign (normal rate) and malignant (higher rate) tumor tissue samples.

Appendix

MASS::biopsy

The dataset includes the following variables:

ID: Sample identifier. **V1** (clump thickness): Thickness of the cell clump. **V2** (uniformity of cell size): Uniformity of cell size. **V3** (uniformity of cell shape): Uniformity of cell shape. **V4** (marginal adhesion): Marginal adhesion. **V5** (single epithelial cell size): Size of individual epithelial cells. **V6** (bare nuclei):

Presence of bare nuclei. **V7** (bland chromatin): Bland chromatin. **V8** (normal nucleoli): Normal nucleoli. **V9** (mitoses): Number of mitoses. **class**: Tumor classification, which can be “benign” or “malignant”.

corrplot

The **corrplot** library is a specialized tool for intuitively and attractively visualizing correlation matrices. It facilitates the interpretation of relationships between variables by providing a variety of correlation plots, such as: **Correlation Diagrams** (Displays the correlation matrix with colors and labels representing the strength and direction of correlations between pairs of variables), **Heatmaps** (Uses colors to visualize the magnitude of correlations, making it easier to identify patterns), among others.

GGally

The **GGally** library in R is an extension of the **ggplot2** package, designed to simplify the visualization and exploratory analysis of data. It provides a series of functions that facilitate the creation of complex and customized plots. Specifically, we will use the **ggcorr** function, which creates **correlation matrices** to help identify patterns and relationships between variables. **In other practices, we have used the corrplot library to create correlation matrices.**

factoextra

The **factoextra** library in R is designed to facilitate the visualization and interpretation of multivariate analysis results, such as Principal Component Analysis (PCA) and Factor Analysis. It includes several useful functions for these tasks: **fviz_pca_biplot** (generates biplots to visualize both observations and variables in the principal component space), **fviz_pca_var** (shows how variables contribute to the principal components), **fviz_pca_ind** (visualizes the distribution of observations in the reduced principal component space), **fviz_eig** (displays a plot of eigenvalues to represent the proportion of variance explained by each component), among others.

gridExtra

gridExtra is an R library that extends the visualization capabilities of the grid graphics system, allowing for flexible combination of multiple plots and tables into a single figure. It provides functions such as **grid.table()** to include tables within plots. It is particularly useful for creating complex compositions of graphs and tables, facilitating the presentation of results in a single visual space.